# AD 2. AERODROMES

## VEBN AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VEBN - LAL BAHADUR SHASTRI AIRPORT, VARANASI

## VEBN AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

| | | | |
|---|---|---|---|
| 1 | Aerodrome reference point coordinates and its site | 252704N 0825131E<br>MAG BRG 99.88 DEG 1051M from<br>physical extremity of RWY 09. | |
| 2 | Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves | 320 DEG/18KM from Varanasi Railway Station | |
| 3 | Aerodrome elevation and reference temperature | 270 FT / 42.0 DEG C | |
| 4 | Magnetic variation, date of information and annual change | 0.25 DEG W (2010) /0.0166 DEG E | |
| 5 | Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available) | Airport Director Airports Authority of India, Lal Bahadur Shastri Airport,<br>Varanasi, Varanasi-221006, | |
| | | Telephone: | (91) 542-2622155 (91) 542-2622081-85<br>(91) 9415223071<br>(91) 542-2622091 (91) 542-2623671<br>(91) 542-2622631 |
| | | Fax: | (91) 542 2622320 |
| | | AFS: | VEBNYHYX |
| | | Email: | apdvns@aai.aero |
| 6 | Types of traffic permitted (IFR/VFR) | IFR/VFR | |
| 7 | Remarks | a. Vertical Datum: EGM 08<br>b. Aerodrome Reference Code: 4C<br>c. Non scheduled aircraft operators are required to file their slot request on apdvns@aai.aero for clearance.<br>d. All non scheduled / General Aviation movements will be subject to positive approval from Airport Director. | |

## VEBN AD 2.3 OPERATIONAL HOURS

| | | |
|---|---|---|
| 1 | Aerodrome Operator | MON-FRI 0400-1230 UTC (0930-1800 IST)<br>SAT,SUN+HOL: NIL |
| 2 | Custom and immigration | AS ATS See Serial No. 12, Remark No. 2 |
| 3 | Health and sanitation | AS ATS |
| 4 | AIS briefing office | As ATSAs ATS |
| 5 | ATS reporting office (ARO) | As ATS |
| 6 | MET Briefing office | AS ATS |
| 7 | Air Traffic Service | H24 |
| 8 | Fuelling | AS ATS |
| 9 | Handling | AS ATS |
| 10 | Security | AS ATS |
| 11 | De-icing | NIL |

| | | |
|---|---|---|
| 12 | Remarks | 1. Customs and immigration facilities are provided on limited basis to cover operations of scheduled intl. Flights. The facilities can be arranged to cover any authorized non-sked operations O/R with 24HRS PN to Airport Director.<br><br>2. Non Sked flights to obtain positive Clearance prior to departure due shortage of parking stands.<br><br>3. The ATS Approved Hourly Runway Traffic Handling Capacity is as Follows:<br>For RWY27<br>Maximum Number of Arrival and Departure- 12<br>Maximum Number of Arrival Only – 08<br>Maximum Number of Departure Only – 10<br><br>For RWY 09<br>Maximum Number of Arrival and Departure- 12<br>Maximum Number of Arrival Only – 06<br>Maximum Number Of Departure Only – 15 |

**VEBN AD 2.4 HANDLING SERVICES AND FACILITIES**

| | | |
|---|---|---|
| 1 | Cargo-handling facilities | Manual - Limited |
| 2 | Fuel and Oil types | Jet A1<br>NIL |
| 3 | Fuelling facilities and capacity | 10 BOWSERS:1 of 27000Ltrs, 7 of16000 Ltrs., 1 of 11000 Ltrs & 1 of<br>6000Ltrs.;13.3Ltrs/Sec |
| 4 | De-icing facilities | NIL |
| 5 | Hangar space for visiting aircraft | NIL |
| 6 | Repair facilities for visiting aircraft | NIL |
| 7 | Remarks | Mooring Facility: NIL |

**VEBN AD 2.5 PASSENGER FACILITIES**

| | | |
|---|---|---|
| 1 | Hotel(s) at or in the vicinity of aerodrome | In the city |
| 2 | Restaurant(s) at or in the vicinity of aerodrome | At AD and in the city |
| 3 | Transportation possibilities | Bus, Taxi, Car hire from AD |
| 4 | Medical Facilities | First aid at AD. Hospital in the city. |
| 5 | Bank and post office at or in the vicinity of aerodrome | Banks: ATM available<br>Post office: Extension Counter AVBL |
| 6 | Tourist office | In the City. |
| 7 | Remarks | NIL |

**VEBN AD 2.6 RESCUE AND FIRE FIGHTING SERVICES**

| | | |
|---|---|---|
| 1 | Aerodrome category for fire fighting | Within ATS HR: CAT-7 |
| 2 | Rescue equipment | Available as per Category |

| | | |
|---|---|---|
| 3 | Capability for removal of disabled aircraft | 1. The critical aircraft for the airport is A321.<br>2. Primary responsibility for removal of disabled aircraft rests with the concerned airline.<br>3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit.<br>4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations:<br>Airport Director, AAI, LBSI Airport, Varanasi.<br>Contact No : +91-542-2622155, +91-9415223071<br>Email : apdvns@aai.aero |
| 4 | Remarks | NIL |

## VEBN AD 2.7 SEASONAL AVAILABILITY CLEARING

| | | |
|---|---|---|
| 1 | Type(s) of clearing equipment | NIL |
| 2 | Clearance priorities | NIL |
| 3 | Remarks | NIL |

## VEBN AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

| | | |
|---|---|---|
| 1 | Designation, surface and strength of aprons | Refer Aircraft Parking/Docking Chart |
| 2 | Designation, width, surface and strength of taxiways | Refer Aircraft Parking/Docking Chart |
| 3 | Location and elevation of altimeter checkpoints | Location At apron Elevation 264 FT |
| 4 | Location of VOR checkpoints | TWY A: 252705.05N 0825119.61E / ELEV 268FT<br>TWY B: 252705.35N 0825114.41E / ELEV 268FT |
| 5 | Position of INS checkpoints | NIL |
| 6 | Remarks | **1. Coordinates of RWY Holding Position**<br>**TWY** — **Coordinates** — **Elevation (FT)**<br>TWY A — 252705.82N 0825119.65E — 267.92<br>TWY B — 252706.10N 0825114.45E — 267.48<br>TWY C — 252706.82N 0825058.76E — 266.56<br>**2. Coordinates of RWY and Taxi Intersection**<br>**TWY** — **Coordinates** — **Elevation (FT)**<br>TWY A — 252708.75N 0825119.83E — 269.03<br>TWY B — 252709.01N 0825114.63E — 268.24<br>TWY C — 252709.78N 0825058.95E — 266.54<br>**3. Vertical Datum: EGM08** |

## VEBN AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

| | | |
|---|---|---|
| 1 | Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands | Taxiing guidance signs at all the intersection with TWY and RWY and at all holding position Guidelines at Apron |
| 2 | Runway and taxiway markings and lights | RWY<br>Markings:Designation, THR, TDZ, Centreline, Edge,<br>Lights:THR, RWY Edge, RWY End<br>TWY<br>Marking:Centreline, Edge Holding Positions,<br>Lights :Edge |
| 3 | Stop bars (if any) | NIL |
| 4 | Remarks | RWY guard lights are provided at each RWY and TWY intersection |

## VEBN AD 2.10 AERODROME OBSTACLES

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| 27/TKOF 09/APCH | OTHER | 252705.1N 0825045.9E | 283 FT | NO | Mobile Road Traffic (Road elev. 81.3 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | OTHER | 252705.0N 0825045.0E | 287 FT | NO | Mobile Road Traffic (Road elev. 82.4 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | BUILDING | 252704.8N 0825041.6E | 293 FT | NO | Building |
| 27/TKOF 09/APCH | TREE | 252705.2N 0825039.1E | 305 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252705.8N 0825040.7E | 310 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | BUILDING | 252706.1N 0825041.1E | 298 FT | NO | Building |
| 27/TKOF 09/APCH | POLE | 252706.3N 0825040.9E | 290 FT | NO | Electric Pole |
| 27/TKOF 09/APCH | TREE | 252706.4N 0825039.2E | 314 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252707.3N 0825036.1E | 311 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252707.9N 0825035.3E | 322 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | BUILDING | 252707.3N 0825041.4E | 290 FT | NO | Building |
| 27/TKOF 09/APCH | TREE | 252706.9N 0825041.3E | 308 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | POLE | 252707.4N 0825042.7E | 288 FT | NO | Electric Pole |
| 27/TKOF 09/APCH | BUILDING | 252708.8N 0825044.9E | 280 FT | NO | Localizer Hut |
| 27/TKOF 09/APCH | OTHER | 252710.2N 0825051.5E | 267 FT | NO | Approach Light |
| 27/TKOF 09/APCH | NAVAID | 252710.3N 0825047.8E | 274 FT | NO | LOC Monitor Antenna (Rod Top - 84.2 m.) |
| 27/TKOF 09/APCH | OTHER | 252710.5N 0825044.7E | 284 FT | NO | Mobile Road Traffic (Road Elev. 81.7 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | OTHER | 252708.8N 0825042.7E | 287 FT | NO | Mobile Road Traffic (Road Elev. 82.4 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | TREE | 252709.3N 0825043.1E | 290 FT | NO | Tree |
| 27/TKOF 09/APCH | OTHER | 252710.6N 0825041.6E | 287 FT | NO | Mobile Road Traffic (Road elev. 82.5 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | TREE | 252709.5N 0825039.1E | 300 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | BUILDING | 252710.9N 0825035.1E | 299 FT | NO | House |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| 27/TKOF 09/APCH | TREE | 252709.6N 0825028.8E | 328 FT | NO | Tree |
| 27/TKOF 09/APCH | TREE | 252711.1N 0825028.2E | 321 FT | NO | Tree |
| 27/TKOF 09/APCH | BUILDING | 252711.7N 0825035.0E | 299 FT | NO | Building |
| 27/TKOF 09/APCH | TREE | 252708.7N 0825036.7E | 310 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252711.5N 0825037.0E | 324 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252713.6N 0825038.9E | 304 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252714.5N 0825039.3E | 322 FT | NO | Tree |
| 27/TKOF 09/APCH | TREE | 252714.4N 0825041.1E | 317 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | BUILDING | 252714.7N 0825038.0E | 292 FT | NO | Building |
| 27/TKOF 09/APCH | TREE | 252715.4N 0825038.1E | 308 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | OTHER | 252716.1N 0825038.9E | 291 FT | NO | Sintex tank on Building (Building top 87.8 m.) |
| 27/TKOF 09/APCH | POLE | 252716.7N 0825038.8E | 297 FT | NO | Electric Pole |
| 27/TKOF 09/APCH | TREE | 252716.9N 0825038.3E | 314 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | OTHER | 252715.9N 0825045.5E | 283 FT | NO | Mobile Road Traffic (Road elev. 81.3 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | OTHER | 252705.1N 0825045.4E | 289 FT | NO | Mobile Road Traffic (Road elev. 83.0 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | TREE | 252705.9N 0825041.2E | 313 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | OTHER | 252706.9N 0825042.3E | 284 FT | NO | Sintex Tank on Building (Building Top 85.6 m.) |
| 27/TKOF 09/APCH | OTHER | 252708.4N 0825044.7E | 282 FT | NO | L/C Rod on LOC Hut |
| 27/TKOF 09/APCH | ANTENNA | 252708.8N 0825044.9E | 282 FT | NO | Antenna on LOC Hut |
| 27/TKOF 09/APCH | OTHER | 252710.6N 0825042.0E | 286 FT | NO | Mobile Road Traffic (Road elev. 82.2 m. + 5 m. Traffic ht.) |
| 27/TKOF 09/APCH | TREE | 252708.1N 0825036.6E | 313 FT | NO | Group of Tree |
| 27/TKOF 09/APCH | TREE | 252709.3N 0825037.8E | 316 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252702.1N 0825246.4E | 303 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | OTHER | 252705.1N 0825234.2E | 269 FT | NO | Approach Light |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| 27/APCH 09/TKOF | POLE | 252658.3N 0825249.9E | 299 FT | NO | Electric Pole |
| 27/APCH 09/TKOF | TREE | 252659.5N 0825310.2E | 339 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252711.5N 0825301.1E | 318 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252710.9N 0825259.3E | 315 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252706.2N 0825252.9E | 302 FT | NO | Tree |
| 27/APCH 09/TKOF | TREE | 252706.8N 0825252.7E | 341 FT | NO | Tree |
| 27/APCH 09/TKOF | POLE | 252705.2N 0825249.8E | 314 FT | NO | Light Pole |
| 27/APCH 09/TKOF | TREE | 252706.6N 0825248.8E | 302 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252708.1N 0825249.7E | 326 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252709.6N 0825252.0E | 306 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252711.0N 0825252.9E | 342 FT | NO | Tree |
| 27/APCH 09/TKOF | TREE | 252710.1N 0825250.5E | 329 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252708.8N 0825248.7E | 328 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252708.2N 0825245.7E | 307 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252706.1N 0825244.8E | 295 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252707.1N 0825245.2E | 308 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | OTHER | 252708.2N 0825241.1E | 281 FT | NO | Mobile Road Traffic (Road Elev. 80.7 m. + 5 m. Traffic ht.) |
| 27/APCH 09/TKOF | POLE | 252709.4N 0825241.1E | 287 FT | NO | Light Pole |
| 27/APCH 09/TKOF | OTHER | 252710.2N 0825241.0E | 284 FT | NO | Mobile Road Traffic (Road Elev. 81.6 m. + 5 m. Traffic ht.) |
| 27/APCH 09/TKOF | POLE | 252710.1N 0825240.2E | 287 FT | NO | Light Pole |
| 27/APCH 09/TKOF | TREE | 252706.9N 0825248.6E | 296 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | TREE | 252710.2N 0825255.6E | 321 FT | NO | Group of Tree |
| 27/APCH 09/TKOF | OTHER | 252652.8N 0825342.0E | 406 FT | NO | Mast |
| 27/APCH 09/TKOF | OTHER | 252650.4N 0825342.7E | 398 FT | NO | Mast |
| In circling area and at AD | TREE | 252658.6N 0825115.9E | 354 FT | NO | Group of Tree |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| In circling area and at AD | ANTENNA | 252700.4N 0825112.9E | 336 FT | NO | VHF Antenna on Flood Light Mast (Flood Mast Top - 98.5 m.) |
| In circling area and at AD | OTHER | 252657.8N 0825108.5E | 411 FT | NO | Mast |
| In circling area and at AD | POLE | 252705.9N 0825108.4E | 273 FT | NO | Spot Light |
| In circling area and at AD | OTHER | 252650.9N 0825048.6E | 438 FT | NO | Mast |
| In circling area and at AD | POLE | 252704.5N 0825044.1E | 298 FT | NO | Electric Pole |
| In circling area and at AD | TREE | 252704.0N 0825043.3E | 294 FT | NO | Tree |
| In circling area and at AD | TREE | 252704.1N 0825041.4E | 311 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252703.6N 0825039.9E | 331 FT | NO | Group of Tree |
| In circling area and at AD | BUILDING | 252704.1N 0825039.2E | 297 FT | NO | House |
| In circling area and at AD | TREE | 252703.3N 0825035.2E | 360 FT | NO | Group of Tree |
| In circling area and at AD | OTHER | 252710.1N 0825053.6E | 267 FT | NO | Approach Light |
| In circling area and at AD | BUILDING | 252717.3N 0825040.3E | 298 FT | NO | Building |
| In circling area and at AD | TREE | 252717.9N 0825046.4E | 339 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252716.6N 0825047.8E | 318 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252716.2N 0825047.9E | 330 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252715.8N 0825047.4E | 284 FT | NO | Light Pole |
| In circling area and at AD | POLE | 252715.6N 0825051.0E | 284 FT | NO | Light Pole |
| In circling area and at AD | TREE | 252716.8N 0825053.9E | 313 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252715.4N 0825056.4E | 284 FT | NO | Light Pole |
| In circling area and at AD | TREE | 252715.8N 0825056.1E | 306 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252716.7N 0825056.3E | 333 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252716.0N 0825057.1E | 312 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252717.4N 0825100.1E | 318 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252715.1N 0825102.6E | 282 FT | NO | Light Pole |
| In circling area and at AD | POLE | 252714.6N 0825113.3E | 282 FT | NO | Light Pole |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| In circling area and at AD | TREE | 252718.1N 0825046.1E | 346 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252718.4N 0825040.5E | 310 FT | NO | Pole on Building (Building Top 90.2 m.) |
| In circling area and at AD | TREE | 252658.4N 0825215.4E | 317 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252659.2N 0825218.5E | 316 FT | NO | Group of Tree |
| In circling area and at AD | NAVAID | 252701.6N 0825219.4E | 288 FT | NO | GP DME (l) |
| In circling area and at AD | BUILDING | 252701.9N 0825219.5E | 276 FT | NO | GP Hut top |
| In circling area and at AD | NAVAID | 252701.8N 0825219.6E | 313 FT | NO | GP Main Antenna (l) |
| In circling area and at AD | NAVAID | 252701.7N 0825222.6E | 288 FT | NO | GP Monitor Antenna (Rod Top - 87.9 m.) |
| In circling area and at AD | OTHER | 252630.8N 0825230.8E | 434 FT | NO | Mast |
| In circling area and at AD | TREE | 252658.7N 0825234.8E | 304 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252658.4N 0825240.2E | 309 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252658.3N 0825242.1E | 311 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252711.5N 0825248.7E | 334 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252713.2N 0825249.2E | 350 FT | NO | Tree |
| In circling area and at AD | OTHER | 252744.0N 0825239.2E | 428 FT | NO | Mast |
| In circling area and at AD | TREE | 252711.1N 0825246.2E | 327 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252710.7N 0825245.2E | 321 FT | NO | Group of Tree |
| In circling area and at AD | TANK | 252710.6N 0825244.9E | 294 FT | NO | Sintex Tank on Building (Building Top 88.9 m.) |
| In circling area and at AD | TREE | 252710.6N 0825241.5E | 299 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252711.9N 0825238.7E | 310 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252711.5N 0825239.2E | 302 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252710.2N 0825236.7E | 286 FT | NO | Light Pole |
| In circling area and at AD | POLE | 252710.3N 0825234.0E | 287 FT | NO | Light Pole |
| In circling area and at AD | TREE | 252712.1N 0825231.8E | 322 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252710.6N 0825228.9E | 286 FT | NO | Light Pole |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| In circling area and at AD | ANTENNA | 252709.5N 0825222.8E | 275 FT | NO | RVR Antenna (MET Pole) |
| In circling area and at AD | ANTENNA | 252709.6N 0825221.7E | 275 FT | NO | RVR Antenna (MET Pole) |
| In circling area and at AD | OTHER | 252710.8N 0825221.8E | 290 FT | NO | WDI |
| In circling area and at AD | TREE | 252712.5N 0825220.2E | 319 FT | NO | Group of Tree |
| In circling area and at AD | BUILDING | 252711.8N 0825218.2E | 296 FT | NO | Building |
| In circling area and at AD | TREE | 252712.7N 0825216.1E | 321 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252712.0N 0825214.6E | 322 FT | NO | Group of Tree |
| In circling area and at AD | BUILDING | 252712.1N 0825213.1E | 290 FT | NO | Building |
| In circling area and at AD | OTHER | 252709.9N 0825222.3E | 276 FT | NO | L/c Rod near RVR Antenna |
| In circling area and at AD | ANTENNA | 252709.6N 0825223.2E | 300 FT | NO | MET Antenna |
| In circling area and at AD | ANTENNA | 252702.9N 0825128.2E | 324 FT | NO | Antenna on Fire Building |
| In circling area and at AD | OTHER | 252702.9N 0825128.0E | 321 FT | NO | L/c Rod on Fire Station Tower (Fire Station Tower Top 94.2 m.) |
| In circling area and at AD | OTHER | 252703.6N 0825127.2E | 282 FT | NO | Fire High Point |
| In circling area and at AD | TREE | 252701.1N 0825136.7E | 314 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252700.7N 0825137.7E | 323 FT | NO | Tree |
| In circling area and at AD | TREE | 252700.9N 0825142.0E | 314 FT | NO | Tree |
| In circling area and at AD | TREE | 252659.2N 0825152.7E | 319 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252658.6N 0825157.7E | 336 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252712.3N 0825210.3E | 318 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252711.8N 0825208.4E | 285 FT | NO | Light Pole |
| In circling area and at AD | TREE | 252712.4N 0825206.2E | 305 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252712.0N 0825204.9E | 286 FT | NO | Light Pole |
| In circling area and at AD | BUILDING | 252712.5N 0825204.8E | 293 FT | NO | Building |
| In circling area and at AD | TREE | 252712.5N 0825203.6E | 302 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252712.7N 0825200.3E | 307 FT | NO | Group of Tree |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| In circling area and at AD | POLE | 252712.4N 0825158.9E | 286 FT | NO | Light Pole |
| In circling area and at AD | TREE | 252714.7N 0825201.4E | 332 FT | NO | Group of Tree |
| In circling area and at AD | TREE | 252715.0N 0825124.8E | 318 FT | NO | Group of Tree |
| In circling area and at AD | POLE | 252714.1N 0825122.3E | 284 FT | NO | Light Pole |
| In circling area and at AD | POLE | 252714.2N 0825119.6E | 285 FT | NO | Light Pole |
| In circling area and at AD | TOWER | 252905.3N 0824943.4E | 440 FT | NO | Tower |
| In circling area and at AD | TOWER | 252843.3N 0824949.1E | 456 FT | NO | Tower |
| In circling area and at AD | TOWER | 252607.2N 0825110.7E | 421 FT | NO | Tower |
| In circling area and at AD | TOWER | 252555.0N 0825135.3E | 444 FT | NO | Tower |
| In circling area and at AD | TOWER | 252500.7N 0825252.4E | 434 FT | NO | Tower |
| In circling area and at AD | OTHER | 252604.2N 0825334.5E | 412 FT | NO | Pylon Mast |
| In circling area and at AD | TOWER | 252744.6N 0825414.7E | 405 FT | NO | Tower |
| In circling area and at AD | TANK | 252754.1N 0825303.1E | 353 FT | NO | Overhead Water Tank |
| In circling area and at AD | TOWER | 252759.0N 0825251.1E | 403 FT | NO | Tower |

## VEBN AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

| | | |
|---|---|---|
| 1 | Name of the associated meteorological office | Varanasi |
| 2 | Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours | H24 |
| 3 | Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts | Lucknow<br>9 HR |
| 4 | Availability of the trend forecast for the aerodrome and interval of issuance | Trend<br>AVBL. Interval of issuance half hour during watch hour and one hour beyond watch hour of tower. |
| 5 | Information on how briefing and/or consultation is provided | NIL |
| 6 | Types of flight documentation supplied and language(s) used in flight documentation | Tabular form English |
| 7 | Charts and other information displayed or available for briefing or consultation | NIL |
| 8 | Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images; | NIL |
| 9 | The air traffic services unit(s) provided with meteorological information | Aerodrome Control Tower, ACC and CNS (AMSS) |

| | | |
|---|---|---|
| 10 | Additional information, e.g. concerning any limitation of service. | i.Wind Panel No DIWE-03 Installed at Varanasi TWR<br>ii.Documents available on request<br>iii.New Transmissiometer (DRISHTI) Installed And Commissioned For RVR Estimation Of Runway 27 |

## VEBN AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

| Designations | TRUE Bearings | Dimensions of RWY (M) | Strength of pavement (PCN) and associated data) and surface of runway and associated stopways | Geographical coordinates for threshold and runway end |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 09 | 93.11 DEG | 2746 x 45 M | 68/F/C/W/T<br>Asphalt | THR:<br>252709.63N<br>0825102.19E<br>RWY END:<br>252705.19N<br>0825232.05E |
| 27 | 273.11 DEG | 2746 x 45 M | 68/F/C/W/T<br>Asphalt | THR:<br>252705.19N<br>0825232.05E<br>RWY END:<br>252710.03N<br>0825053.90E |

| THR elevation and highest elevation of TDZ of precision APP RWY | Slope of runway and associated stopway | Dimensions of stopway (M) | Dimensions of clearway (M) | Dimensions of strips (M) |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| THR: 268.0FT<br>TDZ: 270.0FT | | NIL | NIL | 2866 x 280 M |
| THR: 268.0FT<br>TDZ: 269.0FT | | NIL | NIL | 2866 x 280 M |

| Dimensions of runway end safety areas | Location and description of arresting system (if any) | Existence of an obstacle-free zone | Remarks. |
|---|---|---|---|
| 11 | 12 | 13 | 14 |
| 240M x 90M | | NIL | Geoid Model: EGM08 |
| 212M x 90M | | NIL | Geoid Model: EGM08 |

## VEBN AD 2.13 DECLARED DISTANCES

| RWY Designator | Take-off run available TORA (M) | Take-off distance available TODA (M) | Accelerate distance available ASDA (M) | Landing distance available LDA (M) | Remarks (including runway entry or start point where alternative reduced declared distances have been declared) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 09 | 2746 | 2746 | 2746 | 2514 | THR Displaced by 232M |
| 27 | 2746 | 2746 | 2746 | 2746 | |

## VEBN AD 2.14 APPROACH AND RUNWAY LIGHTING

| Runway Designator | Type, length and intensity of approach lighting system | Runway threshold lights, colour and wing bars | Type of visual slope indicator system | Length of runway touchdown zone lights |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 09 | SALS<br>420 M<br>LIH | Green | PAPI<br>LEFT/3.00 DEG | |
| 27 | SALS<br>420 M<br>LIH | Green | PAPI<br>LEFT/3.00 DEG | |

| Length, spacing, colour and intensity of runway centre line lights | Length, spacing, colour and intensity of runway edge lights | Colour of runway end lights and wing bars | Length and colour of stopway lights | Remarks |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| | 2746 M<br>60 M<br>White<br>LIH | Red | | SALS from beginning of displaced THR RWY09 with three inset LGT |
| | 2746 M<br>60 M<br>White<br>LIH | Red | | NIL |

## VEBN AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

| | | | |
|---|---|---|---|
| 1 | Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any) | ABN | At Tower Building, FLGW&G EV2SEC |
| | | IBN | NIL |
| 2 | Location and lighting (if any) of anemometer/ landing direction indicator; | LDI | Available |
| | | Anemometer | Wind Panel at TWR |
| 3 | Taxiway edge and taxiway centre line lights; | Edge | All TWY |
| | | Centre Line | --- |
| 4 | Secondary power supply including switch-over time; | Secondary Power supply to all lighting at AD.Switch-over time :15 SEC. | |
| 5 | Remarks | NIL | |

## VEBN AD 2.16 HELICOPTER LANDING AREA

| | | |
|---|---|---|
| 1 | Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area | Not Established |
| 2 | TLOF and/or FATO area elevation: | Not Established |
| 3 | TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking; | Not Established |
| 4 | True bearings of FATO; | Not Established |
| 5 | Declared distances available | Not Established |
| 6 | Approach and FATO lighting; | Not Established |
| 7 | Remarks | Not Established |

## VEBN AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

| | | |
|---|---|---|
| 1 | Airspace designation, geographical coordinates and lateral limits | CTR: Circular area centered on 252704N 0825131E within a 30NM radius. |
| 2 | Vertical limits | FL 105 |
| 3 | Airspace classification | D |
| 4 | Call sign and language(s) of the air traffic services unit providing service; | Varanasi Tower / Varanasi Approach, English |
| 5 | Transition altitude | 4000 FT |
| 6 | Hours of applicability | HO |
| 7 | Remarks | Micro light training flight activity 5NM east south east Banaras Hindu University airfield.<br>Lower limit: GND, Upper limit: 1000FT AMSL. |

## VEBN AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

| Service Designation | Call sign | Channel(s) | SATVOICE Number(s), if available |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| APP | Varanasi Approach | 119.000 MHZ | |
| APP | Varanasi Approach | 123.875 MHZ | |
| TWR | Varanasi Tower | 118.100 MHZ | |
| TWR | Varanasi Tower | 119.000 MHZ | |
| ATIS | Varanasi Information | 126.200 MHZ | |
| ACC | Varanasi Control/ Radar | 118.950 MHZ | |
| ACC | Varanasi Control/ Radar | 119.275 MHZ | |
| ACC | Varanasi Control/ Radar | 119.275 MHZ | |
| ACC | Varanasi Control/ Radar | 128.150 MHZ | |
| ALRS | ----- | 121.500 MHZ | |
| SMC | Varanasi Ground | 121.625 MHZ | |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| 5 | 6 | 7 |
| | AS ATS | SDBY |
| | AS ATS | NIL |
| | AS ATS | When Tower and Approach is combined Main Fequancy-118.1 MHz<br>Standby Frequancy-119.0 MHz |
| | AS ATS | SDBY - When Tower and Approach is combined Main Fequancy-118.1 MHz<br>Standby Frequancy-119.0 MHz |
| | AS ATS | NIL |
| | H24 | NIL |
| | H24 | RSR (S) SDBY |
| | H24 | RSR (N) SDBY |
| | H24 | RSR (N) |
| | H24 | NIL |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| | H24 | Main Frequency |

## VEBN AD 2.19 RADIO NAVIGATION AND LANDING AIDS

| Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid | Identification | Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate | Hours of operation, as appropriate; |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| LOC 27<br>CAT I | IVNS | 109.900 MHz | AS ATS |
| GP 27 | IVNS | 333.800 MHz | AS ATS |
| DME ILS 27 | IVNS | CH36X | AS ATS |
| DVOR/DME | BBN | 113.900 MHz<br>CH86X | H24 |

| Geographical coordinates of the position of the transmitting antenna | Elevation of transmitting antenna of DME/ elevation of GBAS reference point | Service volume radius from the GBAS reference point | Remarks |
|---|---|---|---|
| 5 | 6 | 7 | 8 |
| 252710.5N<br>0825044.1E | | | |
| 252701.8N<br>0825220.3E | | | 1. GLIDE ANGLE: 3.0 DEG<br>2. GP COVERAGE RESTRICTED TO 06 DEG ON 90 SIDE AT 10NM AT 2000FT ABV THR |
| 252701.6N<br>0825220.3E | 288 FT | | Vertical Datum: EGM08 |
| 252715.2N<br>0825134.2E | 294 FT | | 1. DME Collocated with DVOR<br>2. Vertical Datum: EGM08 |

## VEBN AD 2.20 LOCAL AERODROME REGULATIONS

### I. PUSHBACK PROCEDURES:

**1. From aircraft stands 3 & 4 (Code C)**

i. Pushback and start-up shall be facing west only.

ii.Pushback shall be abeam stand 5 only.

iii.Taxi out via taxiway A or B.

iv.Aircraft taxiing in via taxiway A shall proceed to stand 1 and 2 only after the aircraft

v.pushing back from stand 3 has completed the pushback.

**2. From aircraft stand 5 (Code C)**

i. Pushback and Start-up shall be facing west only.

ii. Taxi out via taxiway A or B.

**3. From aircraft stands 6 & 7 (Code C/D)**

i.Pushback and start-up shall be facing west only.

ii. Taxi out via taxiway A or B.

**4. From aircraft stand 8 (Code C)**

i. Pushback and start-up shall be facing west only.

ii. Pushback shall not be beyond abeam stand 9.

iii.Taxi out via taxiway A or B.

**5. From aircraft stand 9 (Code C)**

i. Pushback and start-up shall be facing west only.

ii. After Pushback aircraft shall be pulled ahead abeam stand 9.

iii. Start-up shall be approved abeam parking stand 9 only.

iv.Taxi out via taxiway A or B.

**6. From aircraft stand 10 (Code C)**

i. Pushback and start-up shall be facing west only.

ii. After Pushback aircraft shall be pulled ahead abeam stand 9.

iii. Start-up shall be approved abeam parking stand 9 only.

iv. Taxi out via taxiway A or B.

**Note:**

i. Aircraft stands 1, 2 & 11 are power-in and power-out stands.

ii. Pushback shall not be approved to an aircraft until other aircraft taxing on taxilane is clear of the bay of aircraft requesting pushback.

iii. ATC shall mention the direction of pushback in pushback clearances.

iv. Start-up can be given after completion of pushback.

v. 180 degrees turn on apron/taxiways/taxi lane shall not be given/ approved by ATC to aircraft.

vi. Ensuring correct pushback on Taxi lane as per instructions received from ATC is the responsibility of Airline Operator. All aircraft shall be pushed back up to the specified pushback position marking (i.e., the nose wheel of the aircraft shall be at the point where the stand lead in line meets the aircraft stand taxi lane) corresponding to the stand from which pushback was commenced unless otherwise instructed by ATC.

vii. Clearance of aircraft on apron from ground handling equipment, vehicular traffic is the responsibility of ground handling agents and airlines operator.

viii. No person/vehicle shall operate on manoeuvring area without obtaining authorization from Aerodrome Control Tower. In spite of such authorization, entry to Runway or Runway strip shall be subject to further specific authorization by ATC tower.

**II**. No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

**III. Fuel Dumping Area:**

The sector defined by R180 from BBN to R200 from BBN between 30 to 50 NM from BBN at or above FL60 has been designated as Fuel Dumping Area.

## VEBN AD 2.21 NOISE ABATEMENT PROCEDURES

NIL

**VEBN AD 2.22 FLIGHT PROCEDURES**

**I.RADIO COMMUNICATION FAILURE**

**1.INTRODUCTION:**

Radio communication failure (RCF) procedures are prescribed in Para 15.3 of ICAO PANS-ATM DOC 4444, 15th edition 2007. Based on these provisions, following radio communication failure procedures are established, to standardize the actions to be taken by the pilot of arriving and departing aircraft at Varanasi airport.

**2.GENERAL:**

2.1.All Transponder equipped aircraft experiencing Radio Communication Failure shall set transponder to Mode A/C code 7600 as soon as practicable.
(Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required)

**3.ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:**

3.1.In case of arriving aircraft, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as assigned runway. If radio communication failure occurs after the receipt of final /interception turn or approach clearance, the aircraft may continue its approach to land on such Runway.

3.2.In case arriving aircraft has not been advised of any runway for landing, Runway 27 shall be considered as assigned runway for such arrival.

3.3.In case of departing aircraft from Varanasi returning on account of RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.

3.4.Varanasi ATC shall ensure approach and runway lights of the assigned runway in 'SWITCHED ON' position for RCF aircraft irrespective of day/night/visibility/weather conditions. Approach lights in 'SWITCHED OFF' position shall indicate that the approach should be made from the opposite end of runway. Runway and Approach lights on both sides in 'SWITCHED OFF' position shall indicate non- availability of Runway for aircraft experiencing RCF.

3.5.In strong tail wind conditions for assigned Runway, aircraft will carry out a missed approach on assigned runway and after following complete missed approach procedure for such approach, will carry out any of the published instrument approach procedure for the opposite runway.

**4.PROCEDURE FOR ARRIVALS:**

**4.1.ARRIVING AIRCRAFT – STAR- ASSIGNED**
When STAR has been assigned to arrival,

4.1.1Aircraft shall continue on assigned STAR following all level and speed restrictions of STAR, as far as practicable.
4.1.2At the end of the STAR, descend to 2000 FT and make a convenient turn to intercept localizer or final approach track of the published instrument approach procedure for the assigned runway.

**4.2.ARRIVING AIRCRAFT – STAR NOT ASSIGNED**
When STAR has not been assigned to arrival by ATC, aircraft shall

4.2.1.Continue on ATS route, (if flying offset re-join ATS route as soon as practicable), maintaining/descending to cleared flight level and join 'BBN' VOR hold as published.

4.2.2.If higher, descend to 3300 FT in 'BBN' hold. Leave 'BBN' VOR at 3300 FT to carry out any of the published instrument approach procedure for assigned runway.

**4.3.ARRIVING AIRCRAFT BEING RADAR VECTORED**

Aircraft being radar vectored for approach, on experiencing RCF, shall maintain last assigned level and heading for 3 minutes and then take a convenient turn to proceed direct to 'BBN' VOR descending/maintaining last assigned level, or 3300 FT whichever is higher. If higher, descend to 3300 FT in 'BBN' hold.

4.3.1.Leave 'BBN' VOR at 3300 FT to carry out any of the published instrument approach procedure for assigned runway.

**5.PROCEDURE FOR DEPARTURES:**

**5.1.DEPARTURE INTENDING TO CONTINUE TO DESTINATION**

Departing aircraft experiencing RCF and intending to continue to its filed plan destination shall:

5.1.1.Continue on assigned SID or heading climbing to or maintaining cleared level following level restrictions of the assigned SID as far as practicable.

5.1.2.Three minutes after setting Mode A/C code 7600 or reaching cleared Flight level whichever is later:

i.If following 'SID', continue on SID to join flight plan route and climb to filed Flight level and continue as per the filed flight plan.

ii.If following Radar heading, turn to join flight plan route by shortest route. Joining ATS route, climb to filed Flight level and Continue as per the filed flight plan.

**5.2DEPARTURE INTENDING TO LAND BACK AT VARANASI:**

Departing aircraft experiencing RCF and intending to land back at VARANASI airport shall

5.2.1Continue on Radar heading or assigned SID, climbing to or maintaining the cleared Flight level. However, in case the RCF is experienced below FL150 and the cleared flight level is above FL150, aircraft shall maintain FL150.

5.2.2 One minute after setting Mode A/C code 7600 or reaching the appropriate level as given in 5.2.1, whichever is later; take a turn to proceed direct to 'BBN' VOR maintaining level.

5.2.3Descend to 3300 FT in 'BBN' hold. Leave 'BBN' VOR at 3300 FT to carry out any of the published instrument approach procedure for assigned runway.

**NOTE:** The provisions relating to SIDs and STARs in Para 4 and para 5 of the RCF procedure shall become applicable on the date of publication of PBN SIDs and STARs for Varanasi airport.

**II**. ADS-B based ATS Surveillance services are provided to suitably equipped aircraft on opportunity basis within the coverage area of Varanasi ADS-B sensors in the jurisdiction of Varanasi Area control.

ADS-B equipped aircraft flying within Varanasi Area Control to comply with relevant provisions.

**III.** Daily between 0400-1200 survellance based approach control service within the jurisdiction of Varanasi Appproach Control shall be provided on VHF 123.875 MHZ (MAIN) and 119.0 MHZ (STANDBY).

The jurisdiction of approach surveillance unit at Varanasi is defined as follows:

a. WI 30 NM from ARP ground to FL 145.

b. BTN 30 NM and 50 NM from ARP FL105 to FL 145 excluding the overlapping portion of airspace of VIR146B/ VIR146A (Prayagraj) and VER75A (Gorakhpur). SSR Transponder and/or ADS-B out equipped aircraft flying within Varanasi approach jurisdiction shall comply with the provisions contained in AIP INDIA ENR 1.6 ATS Surveillance services.

IV. LOW VISIBILITY PROCEDURE

1. BACKGROUND:

To comply with DGCA Civil Aviation Requirements (CAR), Section 8, Series 'C', Part-I on All-weather Operations, Low Visibility Procedure are required at aerodromes for the purpose of ensuring safe operations during CAT II and CAT III approaches and/or low visibility take-offs (LVTO).

As per DGCA Civil Aviation Requirements (CAR), Section 8, Series 'C', Part-I on All-weather Operations "Quote"

An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550m RVR/800 m visibility unless Low Visibility Procedure are enforced. "Unquote".

This provision necessitated the need of Low Visibility Procedure for accommodating/permitting departures in Visibility/RVR less than 800M/550M even at such airports where there are no CAT II and CAT III operations. Accordingly, Low Visibility Procedure has been developed for Varanasi airport to accommodate/permit departures in Visibility/RVR less than 800M/550M.

2. DEFINITIONS AND ABBREVIATIONS:

2.1 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Note: As per para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550M RVR/800M Visibility unless low visibility procedures are enforced.

2.2 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons

2.3 Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

2.4 Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a) Take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.

b) landing in 2 D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

d) Aerodrome operating minima for Airports shall be established by the airline/aircraft operators on the OCA/OCH or DA/DH and ALS facility published by AAI.

e) Adherence to the minima is the responsibility of the pilot in command. However, ATC is responsible to pass the latest meteorological information and status of aeronautical ground lights and other related facilities at the airport to the pilot in command, including any significant changes.

2.5 Visibility - Visibility for aeronautical purposes is the greater of:

a) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background.

b) The greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.

Note 1. — The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2. — The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

2.6 Abbreviations:

ADC : Aerodrome Control

ARFF : Airport Rescue and Fire Fighting Services

ATC : Air Traffic Control

ATM : Air Traffic Management

CFT : Crash Fire Tender

DATIS : Datalink Automatic Terminal Information Service

DG : Diesel Generating Set

LVP : Low Visibility Procedures

MET : Meteorology

RWY : Runway

RVR : Runway Visual Range

SMC : Surface Movement Control

SP : Safeguarding Procedures

TDZ : Touchdown Zone

TWR SUP : Tower Supervisor

TWY : Taxiway

WSO : Watch Supervisory Officer

3. GENERAL:

The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

3.1 Varanasi Airport is equipped with CAT-I ILS system for Runway 27 and Simple Approach Lighting System (SALS) of 420M for Runway 09 and Runway 27.

4. MINIMUM REQUIREMENTS:

The following aeronautical Ground lights and RVR equipment shall be serviceable to the required standard to support Low Visibility Procedures.

a) Runway edge lights,

b) Runway Guard lights,

c) Runway end lights,

d) Real time TDZ RVR.

e) Standby Power supply to maintain switch over time of one Second for Runway Edge Lights and Runway End Lights.

Note 1: To meet the requirement in para 4 (d), Instrument RVR reporting system is available for departure Runway. However, in case RVR equipment is Unserviceable, manual RVR will be taken for departure Runway and at least one manual RVR will be taken before allowing the aircraft to enter runway for departure.

Note 2: At Varanasi, the power supply to Runway Edge Lights and Runway End Lights is provided through online UPS (Uninterrupted power supply). Hence, switch over time is not applicable.

4.1 Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP.

| **Aeronautical Ground Lighting Facility** | **Un-serviceability** | **Restrictions** |
|---|---|---|
| Runway Edge lights | More than 15% of all lights are unserviceable | LVP operations will be suspended. |
| | Any two consecutive lights or more are unserviceable | |
| | Any one of the two circuits serving the power supply goes unserviceable | |

| | | |
|---|---|---|
| Runway End lights | More than 15% of all lights are unserviceable | LVP operations will be suspended. |
| | Any two consecutive lights or more are unserviceable | |
| Standby Generators | Any of the generators is unserviceable | LVP operations will be suspended. |

4.2 Unserviceability of Aeronautical Ground Lights/ Equipment after Implementation of LVP.

When any of the light/equipment mentioned in Para 4 above becomes un-serviceable or fails to meet the required standard during periods of LVP, Duty Officer Tower shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.

Low Visibility Procedure will not be implemented when any of the light/equipment mentioned in Para 4 above is unserviceable or is not maintained as per the required standard.

5. SAFEGUARDING PROCEDURES (SP):

Safeguarding Procedures are instructions for all concerned departments to prepare ground services and facilities for low visibility operations in order that when LVP is implemented all Safeguarding procedures are complete. The Duty Officer TWR in consultation with other relevant units shall give confirmation to WSO that SP is complete before implementation of Low Visibility Procedure.

5.1 Safeguarding Procedures shall be initiated when:

a) When either Visibility or RVR is less than 1200M and visibility/RVR is forecast to deteriorate to 800M or less; and/or

b) The cloud ceiling is less than 400 FT and forecast to fall to 200 FT or less.

5.2 Safeguarding Procedures include:

a) Runway inspection, which shall be carried out by ATCO along with electrical engineers on duty and shall not be carried out by the fire staff.

b) One CFT & one ambulance will be positioned at the road holding point on the road leading to the Runway from New Fire Station and; one CFT & one ambulance will be positioned at the road holding point on the road leading to the Runway from Satellite Fire Station.

c) Stopping of all works on the manoeuvring area and the associated strip area as well as removal of all men and mobile equipment from the said area.

d) All Aerodrome Ground Lights may continue to operate on the main power supply during Safeguarding Procedures.

e) The relevant Aeronautical ground lights must have been inspected during the hour preceding the implementation of LVP, and thereafter once every two-hour period by an electrical engineer. These lighting inspections should be accorded priority.

f) Duty Officer TWR shall check the serviceability of DG set with Electrical wing. Availability of serviceable DG set is mandatory for implementation of LVP.

6. LOW VISIBILITY PROCEDURE:

6 .1 Implementation of Low Visibility Procedure:

Duty Officer Tower shall coordinate with all the agencies to confirm whether the Safeguarding procedures have been completed or not. When Visibility/RVR falls below 800M/550M and/or Cloud Ceiling is 200FT or below and safeguarding procedures are complete, Tower will implement Low Visibility Procedure. Duty Officer Tower shall inform Operations / ARFF, Electrical and Civil department of the imposition of Low Visibility Procedure.

Note: The Low Visibility Procedure shall only be initiated / implemented when a specific request is received from airlines / pilot to conduct take-off in low visibility.

6.2 Action by various units during LVP:

a) Duty MET Officer shall keep Duty Officer Tower informed of any change in Visibility/ RVR.

b) The Duty MET Officer shall ensure that the RVR instruments are serviceable and direct readings of RVR observations are provided at control positions.

c) Whenever RVR instruments are unserviceable, or the take-off is to be conducted in Low Visibility Procedure on Runways not equipped with RVR Instruments, Duty MET Officer shall deploy suitable manpower to obtain Manual RVR.

d) Duty MET Officer shall keep Duty Officer Tower/Tower Supervisor informed of any change in Visibility/ RVR.

e) Duty Officer Tower shall ensure that the towing of aircraft is done under escort of “Follow Me” vehicles. “Follow Me” shall follow the route cleared by ATC.

f) The number of vehicles on the maneuvering area shall be restricted to bare minimum.

g) Duty Officer Tower shall not permit any ground run on the maneuvering area except idle power run on the stands.

h) Duty Officer Tower shall permit towing of only one aircraft at a time on the Apron/Maneuvering area

i) The following may be included in ATIS “Low Visibility Procedure implemented”.

j) Duty Officer Tower shall permit departures only from the beginning of the Runway.

k) Duty personnel in Powerhouse shall continuously monitor the Standby Power supplies and report any unserviceability to Tower immediately.

7. TERMINATION OF LOW VISIBILITY PROCEDURE:

a) When Visibility/RVR improves to 800M/550 M or more and cloud ceiling is 200 feet or Higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP.

b) Duty Officer Tower will intimate Operations / ARFF/ In-Charge Electrical- Engineering regarding the termination of LVP operations.

c) Upon cancellation of LVP, the following message shall be included in two subsequent ATIS broadcasts. “LOW VISIBILITY PROCEDURES CANCELLED”.

d) If Safeguarding Procedures are implemented and LVP are not subsequently implemented, the visibility/RVR improves to more than 1200m and/or the cloud ceiling is 400 FT or higher and both are forecast to remain above the required SP criteria, Duty Officer Tower may cancel Safeguarding Procedures.

8. ACTIONS BY OTHER AGENCIES (AIRLINES, REFUELING COMPANIES, CATERING AGENCIES, ETC.)

a) Every year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

b) All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility operations.

c) A refresher program for ATCO’s and personnel responsible for airside operations shall be conducted every year.

d) All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

e) All the vehicles must have their obstruction lights “ON” during Low Visibility Procedure operations.

## VEBN AD 2.23 ADDITIONAL INFORMATION

1. ADS-B Ground Equipment Commissioned and Operational.

2. Indra Automation System commissioned and operational w.e.f 04th MAY 2017 .

3. Following intersection Take Off Run Available (TORA):

| RWY | From TWY Intersection | TORA (M) |
|---|---|---|
| RWY 09 | TWY A | 2020 |
| RWY 09 | TWY B | 2166 |
| RWY 09 | TWY C | 2605 |
| RWY 27 | Intermediate Turn Pad | 2207 |

4. RWY turn pad capable of handling B777-300 ER type of ACFT, provided at both ends of RWY 09/27
For special OPS with prior permission. PCN TURN PAD RWY 09: 64/F/C/W/T. PCN TURN PAD RWY 27: 68/F/C/W/T.

## VEBN AD 2.24 CHARTS RELATED TO AN AERODROME

1 Aerodrome Chart (17 April 2025)
2 Aircraft Parking / Docking Chart
3 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 09(17 April 2025)
4 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 27(17 April 2025)
5 Aerodrome Obstacle Chart - Type B(17 April 2025)
6 Instrument Approach Chart - ILS (Z) Procedure Rwy 27 (6 AUG 2026)
7 Instrument Approach Chart - VOR Procedure Rwy 09 (6 AUG 2026)
8 Instrument Approach Chart - VOR Procedure Rwy 27 (6 AUG 2026)
9 ATC Surveillance Minimum Altitude Chart (6 AUG 2026)
10 Instrument Approach Chart - RNP Y Procedure RWY 09 (6 AUG 2026)
11 Coding Table - RNP Y Procedure RWY 09
12 Instrument Approach Chart - RNP Y Procedure RWY 27 (6 AUG 2026)
13 Coding Table - RNP Y Procedure RWY 27

## VEBN AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

| Procedure | Procedure Minima | VSS Penetration |
|---|---|---|
| 1 | 2 | 3 |
| | | |